Hypomagnesemia, Seizures, and Intellectual Disability 1 (HOMGSMR1): A Comprehensive Disease Characterization

Disease: Hypomagnesemia, Seizures, and Intellectual Disability 1 (HOMGSMR1 / HSMR) **Causal gene: CNNM2** (Cyclin M2; cyclin and CBS domain divalent metal cation transport mediator 2; HGNC:2020; NCBI Gene 54805; locus 10q24.32) **Identifiers:** MONDO:0020787 · OMIM:616418 · Orphanet (hypomagnesemia–seizures group) **Category:** Mendelian monogenic disorder

> **Evidence-source key:** [H] human clinical/genetic · [M] model organism · [V] in vitro/biochemical/structural · [C] computational. PMIDs cited throughout. This report concerns the **CNNM2**-related disorder, *not* the similarly named TRPM6- or CNNM4-related magnesium disorders.

Summary

Hypomagnesemia, Seizures, and Intellectual Disability 1 (HOMGSMR1) is a rare Mendelian neurodevelopmental and renal electrolyte disorder caused by pathogenic variants in **CNNM2** (Cyclin M2), a magnesium-handling protein expressed on the **basolateral membrane of the renal distal convoluted tubule (DCT)**. CNNM2 contains intracellular **CBS/Bateman domains** that sense cytosolic **Mg-ATP**, and it regulates cellular Mg^{2+} flux—both basolateral extrusion in the kidney and, in cells, uptake through the TRPM7 Mg^{2+}-permeable channel. Loss of CNNM2 function impairs renal Mg^{2+} reabsorption (producing renal Mg^{2+} wasting and hypomagnesemia) and disrupts brain development, yielding the disease's core triad: **hypomagnesemia, seizures, and intellectual disability/developmental delay** [H, PMID: 21397062; PMID: 24699222].

The disorder shows an **allelic-dose (genotype–phenotype) gradient**. Heterozygous, frequently *de novo*, missense or truncating variants cause the classic autosomal-dominant disorder with moderate intellectual disability; biallelic/recessive variants cause a more severe phenotype with brain malformation and severe intellectual disability, and in some cases the

neurological phenotype occurs even without overt hypomagnesemia [H, PMID: 24699222; PMID: 34490037]. Population-genetics constraint data strongly reinforce a haploinsufficiency mechanism: in gnomAD v2.1.1, CNNM2 is extremely loss-of-function intolerant (pLI ≈ 1.0, LOEUF = 0.29, missense Z = 5.77).

This report integrates five convergent layers of evidence—human genetics, population genetics, structural biology, cell biology, and animal models—into a coherent mechanistic model. Management is supportive: **oral/intravenous/subcutaneous magnesium supplementation** plus **antiseizure therapy**; no curative or gene-directed therapy exists. The disorder is ultra-rare (only ~24 pediatric HSMR cases reported by 2022; largest cohort ~34), which constrains firm epidemiological, prognostic, and treatment-outcome estimates.

Key Findings

1. **CNNM2 is the causal gene.** Positional/candidate cloning in two unrelated families with dominant hypomagnesemia identified CNNM2 variants [PMID: 21397062]: *"we identified CNNM2 as a gene involved in renal Mg(2+) handling in patients of two unrelated families with unexplained dominant hypomagnesemia."* CNNM2 localizes to the basolateral membrane of distal tubular segments.
2. **The CBS/Bateman domains sense Mg-ATP and are essential for Mg^{2+} handling.** CBS-domain deletion abolishes Mg^{2+} efflux; a disease-associated CBS substitution *"abrogates Mg(2+) efflux"* and ATP binding [PMID: 24706765].
3. **A structural allosteric switch is disrupted by T568I.** Nucleotide binding triggers a *"twisted towards a flat disc-like"* CBS-module conformation; the dominant mutation **T568I mimics the nucleotide-bound state** [PMID: 25184538].
4. **CNNM2 regulates TRPM7-mediated Mg^{2+} uptake.** CNNM2 *"increases cellular Mg2+ uptake in HEK293 cells … through regulation of the Mg(2+)-permeable cation channel TRPM7"*; mutants fail to do so [PMID: 24699222].
5. **Allelic-dose genotype–phenotype gradient.** *"patients with recessive CNNM2 mutations suffer from brain malformations and severe intellectual disability,"* whereas heterozygous *de novo* missense patients have moderate disability [PMID: 24699222].
6. **CNNM2 is extremely LoF-intolerant** (gnomAD v2.1.1: pLI ≈ 1.0, LOEUF = 0.29, 12 observed vs 66.9 expected LoF, mis_Z = 5.77), supporting haploinsufficiency [C].

7. **Animal models recapitulate both axes.** Mouse *Cnnm2* KO is embryonic-lethal (homozygous) and hypomagnesemic (heterozygous/kidney-specific); zebrafish knockdown produces disturbed brain development [PMID: 33859252; PMID: 28033128; PMID: 24699222].

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1. Disease Information

Overview. HOMGSMR1 (also "Hypomagnesemia, Seizures, and Mental Retardation syndrome," HSMR) is a rare autosomal-dominant (occasionally recessive) Mendelian neurodevelopmental and renal-electrolyte disorder. Loss-of-function/dominant-negative variants in **CNNM2** impair basolateral Mg^{2+} extrusion in the renal DCT, producing renal Mg^{2+} wasting (serum Mg^{2+} <0.7 mmol/L) combined with seizures and intellectual disability/developmental delay [H, PMID: 21397062; PMID: 33600043].

Key identifiers.

Resource	Identifier
MONDO	MONDO:0020787
OMIM (phenotype)	616418
OMIM (gene)	607803
Gene / HGNC	CNNM2 / HGNC:2020
NCBI Gene	54805
Ensembl	ENSG00000148842 (canonical ENST00000369878)
UniProt	Q9H8M5
Locus	10q24.32 (chr10:102,918,294–103,090,222, GRCh38)
ICD-10 / ICD-11	E83.4 / 5C64.4 (disorders of magnesium metabolism — no CNNM2-specific code)
MeSH (closest)	Magnesium Deficiency (D008275); Intellectual Disability (D008607); Seizures (D012640)

Synonyms: HOMGSMR1; HSMR syndrome; CNNM2-related hypomagnesemia; Hypomagnesemia, seizures, and impaired intellectual development 1; Cyclin M2 deficiency.

Information source: A mix of **individual-patient case reports and small cohorts** (landmark multi-family studies Stuiver 2011 [PMID 21397062], Arjona 2014 [PMID 24699222], the largest cohort Franken 2021 [PMID 33600043]) plus mechanistic/biochemical/animal studies—and **aggregated disease-level resources** (OMIM, Orphanet, gnomAD). There is no large population registry or EHR-scale dataset.

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2. Etiology

Primary cause (genetic). Pathogenic variants in **CNNM2**—there is no infectious or environmental cause. The gene was originally identified in two unrelated families with dominant hypomagnesemia [H, PMID: 21397062].

Genetic risk factors. Causal variants are heterozygous (frequently *de novo*) missense and truncating variants, with biallelic variants in severe recessive forms (see §4, §9). The 10q24 locus also carries GWAS signals for serum Mg^{2+} concentration and for schizophrenia (zebrafish *cnnm2* mutants show altered forebrain development, [PMID 30929901])—these are quantitative-trait/susceptibility associations distinct from the Mendelian disorder. CNNM2's extreme LoF intolerance (pLI ≈ 1.0) indicates heterozygous LoF alleles are under strong negative selection.

Environmental / lifestyle factors. None causal. Acquired hypomagnesemia (not HOMGSMR1) is driven by alcoholism, type-2 diabetes, proton-pump inhibitors, calcineurin inhibitors, and thiazide diuretics [H, PMID: 38871680]; these can exacerbate but not cause the syndrome.

Protective factors. No validated protective alleles. Dietary/supplemental Mg^{2+} is therapeutically mitigating but does not fully correct the neurological phenotype.

Gene–environment interactions. Dietary Mg^{2+} intake and Mg^{2+}-wasting medications modulate the degree of hypomagnesemia on top of the genetic lesion; no formal GxE model exists for CNNM2.

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3. Phenotypes

Core triad plus associated features (frequencies qualitative; cohorts small, N≈24–34) [H, PMID: 35170241; PMID: 33600043].

Phenotype	Type	Onset	Frequency	HPO term
Hypomagnesemia (serum Mg^{2+} <0.7 mmol/L)	Lab abnormality	Neonatal/ infantile–childhood	Very frequent (core; occasionally absent)	HP:0002917
Seizures / epilepsy (can be intractable)	Clinical sign	Neonatal–infancy	Frequent (core)	HP:0001250
Intellectual disability / developmental delay	Behavioral/ cognitive	Childhood	Frequent (core)	HP:0001249; HP:0001263
Speech / language delay	Behavioral	Childhood	Common	HP:0000750
Motor impairment / delay	Clinical sign	Infancy–childhood	Common	HP:0001270
Brain malformation	Imaging	Congenital	Subset (can occur w/ o hypomagnesemia)	HP:0002011
EEG abnormalities	Electrophysiology	Variable	Common with seizures	HP:0002353
Congenital heart disease (ASD), sinus arrhythmia	Clinical sign	Congenital	Reported subset	HP:0001631; HP:0011675
Hepatic (liver function) dysfunction	Lab abnormality	Variable	Single reports	HP:0001410

Key quotes: *"only 24 children with HSMR have been reported, mainly manifested as hypomagnesemia, mental retardation, seizures, and language and motor impairment"* [PMID: 35170241]. Neurological disease can occur **without** overt hypomagnesemia: *"CNNM2 Mutations Cause Damaged Brain Development and Intractable Epilepsy in a Patient Without Hypomagnesemia"* [PMID: 34490037].

Severity/progression. Variable expressivity; seizures may be intractable; intellectual disability is typically stable (non-progressive) but lifelong; hypomagnesemia is chronic.

Quality-of-life impact. Intellectual disability, epilepsy, and speech/motor impairment substantially reduce daily functioning, independence, and schooling; chronic Mg^{2+} supplementation causes GI side effects (diarrhea). No disease-specific EQ-5D/SF-36 data exist.

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4. Genetic / Molecular Information

Causal gene. CNNM2 (OMIM 607803; HGNC:2020; UniProt Q9H8M5) encodes Cyclin M2, a ~875-aa integral membrane protein with an extracellular N-terminus and intracellular C-terminus; the ~64-aa N-terminal signal peptide is cleaved and the protein is N-glycosylated at Asn-112, stabilizing it at the plasma membrane [V, PMID: 22399287]. It contains a DUF21 transmembrane domain, a cytosolic **CBS/Bateman module** (residues ~429–584) that binds Mg-ATP and undergoes a twisted→flat switch [V, PMID: 25184538], and a cyclic-nucleotide-binding-homology domain. Two isoforms (CNNM2a/CNNM2b) heterodimerize [V, PMID: 22399287].

Pathogenic variants. Predominantly **heterozygous *de novo* missense or truncating**, with biallelic genotypes in severe cases:

Variant (cDNA / protein)	Type	Classification	Inheritance	Reference
p.Thr568Ile (T568I)	Missense (CBS/ Bateman)	Pathogenic (founding dominant variant)	Dominant	PMID: 25184538
c.1448delT (p.Val483GlyfsTer29)	Frameshift/ truncating	Pathogenic (ACMG), *de novo*	Dominant	PMID: 37532502
c.566A>G (p.Tyr189Cys)	Missense	Pathogenic (ACMG), *de novo*, parents WT	Dominant	PMID: 35170241
p.E298del	In-frame deletion	Novel, pathogenic (functional study)	—	PMID: 40612795
p.P360R	Missense	Novel, pathogenic (functional study)	—	PMID: 40612795
11 novel variants (cohort)	Mixed	Across 9 sporadic + 2 families	Mixed	PMID: 33600043

Classification (ACMG/AMP): pathogenic/likely pathogenic, commonly using PVS1 (truncating), PS2 (*de novo*), PM2 (absent/rare in gnomAD), PP4 (phenotype match) [H, PMID: 37532502].

Allele frequency / constraint (gnomAD v2.1.1, queried via GraphQL API): pathogenic variants are absent/ultra-rare (PM2-supporting). CNNM2 is extremely LoF-intolerant: **pLI = 0.99999998 (≈1.0)**, **oe_lof = 0.179 (90% CI 0.115–0.291; LOEUF = 0.29)**, only **12 observed vs 66.9 expected LoF**, **missense Z = 5.77**, synonymous Z = −1.47 (unconstrained). A LOEUF of 0.29 places CNNM2 in the most-constrained decile, consistent with haploinsufficiency and dominant (often *de novo*) inheritance [C, finding F009].

Origin: germline; frequently *de novo*. No somatic-cancer disease role for the syndrome (though CNNM–PRL interactions are oncogenic in tumor biology—see §6).

Functional consequence: loss of function (reduced Mg^{2+}-sensitive Na^{+} currents; abolished Mg^{2+} efflux for CBS-domain variants) [V, PMID: 21397062; PMID: 24706765]; dominant-negative action is plausible given heterodimerization and dominant inheritance.

Functional partners / modifier genes. TRPM7—CNNM2 promotes cellular Mg^{2+} uptake through regulating TRPM7; mutant CNNM2 fails to do so [V, PMID: 24699222]. Co-functioning DCT transporters (apical TRPM6; basolateral SLC41A3) and the transcriptional regulator PCBD1 could plausibly modify phenotype [H, PMID: 38871680; PMID: 26446763]. No formal clinical modifier gene is established.

Epigenetic / chromosomal. No disease-specific methylation/histone signature or characteristic chromosomal abnormality; HOMGSMR1 is a single-gene disorder.

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5. Environmental Information

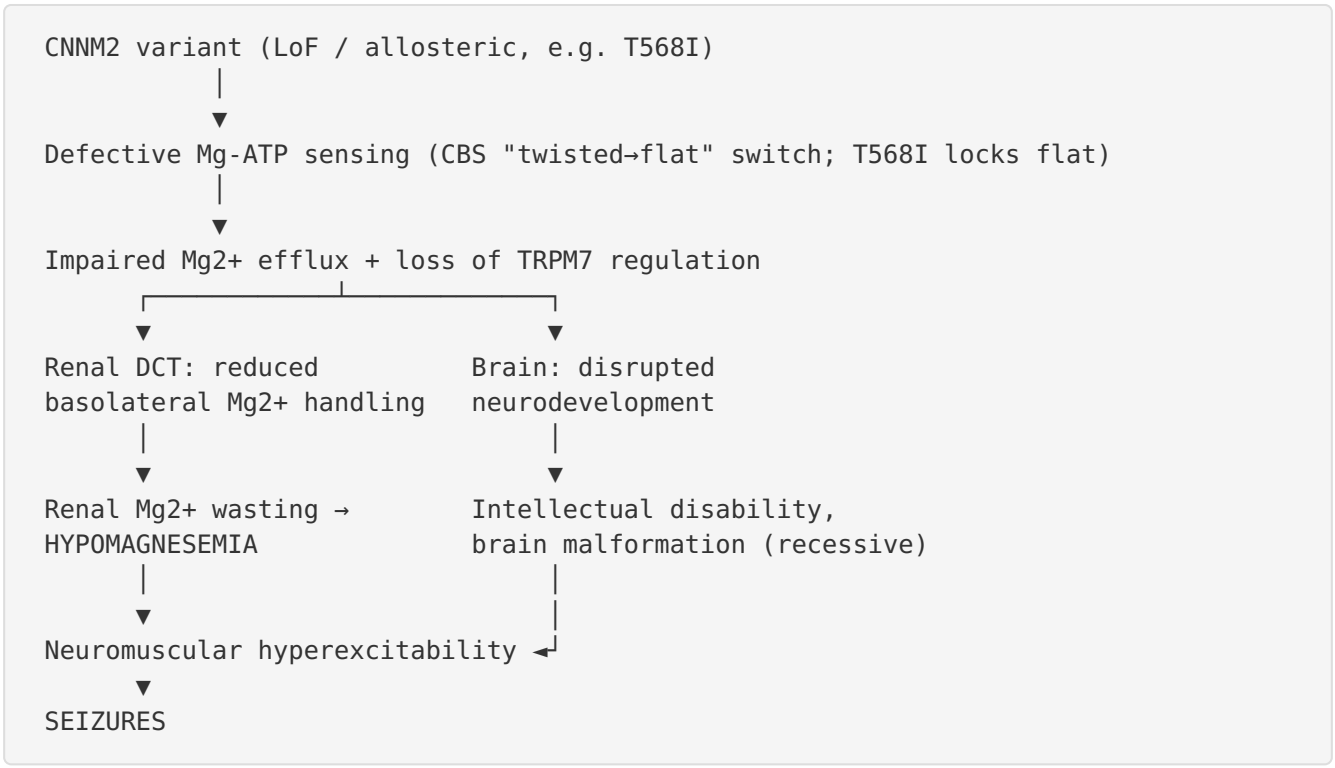
- **Environmental/toxin factors:** none causal for HOMGSMR1.
- **Lifestyle:** dietary Mg^{2+} intake modulates severity; Mg^{2+}-depleting behaviors/conditions worsen hypomagnesemia [H, PMID: 38871680].
- **Infectious agents:** none; non-infectious genetic disorder.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A germline **CNNM2 variant** (CBS/DUF21 missense or truncating) **leads to** a structurally/functionally defective Cyclin M2 protein (reduced stability / disrupted Mg-ATP binding) [V/C, PMID: 22399287]. *(Demonstrated.)*
2. The defective CBS module **results in** loss of Mg^{2+}-dependent ATP sensing and distortion of the twisted↔flat allosteric switch (T568I locks the flat/nucleotide-bound state): *"the substitution of one amino acid residue in the CBS domains of CNNM2 … abrogates Mg(2+) efflux"* [V, PMID: 24706765; PMID: 25184538]. *(Demonstrated — crystallography.)*
3. This **leads to** impaired basolateral **Mg^{2+} extrusion** in DCT epithelial cells (diminished Mg^{2+}-sensitive Na^{+} currents in mutants) and loss of regulation of **TRPM7-mediated uptake** [V, PMID: 21397062; PMID: 24699222]. *(Demonstrated.)*
4. Reduced DCT Mg^{2+} handling **results in** decreased transcellular Mg^{2+} reabsorption (the nephron's final fine-tuning step) → renal **Mg^{2+} wasting** → systemic **hypomagnesemia** [H, PMID: 21397062; PMID: 38871680]. *(Demonstrated.)*
5. **Branch A (systemic):** Hypomagnesemia **leads to** neuronal/neuromuscular hyperexcitability (loss of NMDA-receptor Mg^{2+} block; membrane destabilization) → **seizures**, tetany, cardiac arrhythmia [H, PMID: 21397062]. *(Demonstrated/clinical.)*
6. **Branch B (Mg^{2+}-independent neurodevelopmental):** CNNM2 is expressed in brain and required for normal forebrain/neurodevelopment; loss **leads to** abnormal brain development and intellectual disability/epilepsy **even when serum Mg^{2+} is normal** [H/M, PMID: 34490037; PMID: 30929901]; zebrafish *cnnm2* knockdown causes disturbed brain development [M, PMID: 24699222]. *(Inferred in humans; demonstrated in zebrafish.)*



Supporting detail

- **Molecular pathways / biochemistry:** The CBS/Bateman module binds **ATP (not AMP)** **Mg^{2+}-dependently**; *"Mg(2+) favours ATP binding by alleviating the otherwise negative charge repulsion existing between acidic residues and the polyphosphate group of ATP"* [V, PMID: 25184538]. Activity is regulated by **PRL (phosphatase of regenerating liver)** phosphatases binding the Bateman module, raising intracellular Mg^{2+} (oncogenic in cancer); the PRL-1·CNNM2 structure is solved and a **phosphocysteine** intermediate mediates the switch [V, PMID: 27899452; PMID: 27856537].
- **Protein dysfunction / structure:** Nucleotide binding triggers *"a conformational change in the CBS module from a twisted towards a flat disc-like structure"*, and *"the T568I mutation, which causes dominant hypomagnesaemia, mimics the structural effect induced by nucleotide binding"* [V, PMID: 25184538].
- **Transport directionality nuance:** CNNM2 is reported both as a basolateral Mg^{2+} **extruder** [V, PMID: 21397062] and as promoting **uptake via TRPM7** [V, PMID: 24699222]; whether it is a direct transporter or an Mg^{2+} sensor/regulator remains debated [PMID 33859252]. All data converge on loss of CNNM2-dependent Mg^{2+} handling as the pathogenic event.

- **Metabolic relevance:** Mg^{2+} is a cofactor for >600 enzymes as Mg-ATP, so deficiency broadly impairs energy metabolism, muscle contraction, and neurotransmission [H, PMID: 38871680].

GO terms: GO:0015693 (magnesium ion transport); GO:0010960 (magnesium ion homeostasis); GO:0015095 (magnesium ion transmembrane transporter activity); GO:0005524 (ATP binding); GO:0007420 (brain development). **CL terms:** CL:1000849 (kidney distal convoluted tubule epithelial cell); CL:0000540 (neuron). **CHEBI:** CHEBI:18420 (magnesium(2+)); CHEBI:30616 (ATP).

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7. Anatomical Structures Affected

- **Organ level — primary:** kidney (distal convoluted tubule; UBERON:0001292 nephron tubule, UBERON:0005097/UBERON:0004135 distal convoluted tubule) and brain (UBERON:0000955). Secondary/associated: heart (ASD, arrhythmia; UBERON:0000948), liver (UBERON:0002107; single reports).
- **Body systems:** urinary/renal, central nervous, cardiovascular, endocrine-metabolic.
- **Tissue/cell level:** renal **DCT epithelial cells** (CNNM2 *"predominantly found along the basolateral membrane of distal tubular segments involved in Mg(2+) reabsorption"* [PMID: 21397062]); CNS **neurons** (forebrain/pallium, hypothalamus, tectum in zebrafish [PMID: 30929901]).
- **Subcellular level: basolateral plasma membrane** (GO:0016323); **endoplasmic reticulum** (signal-peptide cleavage/processing, GO:0005789); cytosolic CBS-module interface [V, PMID: 22399287].
- **Laterality:** bilateral (systemic renal wasting and generalized CNS involvement).

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8. Temporal Development

- **Onset:** congenital to neonatal/infantile; seizures often in the first months of life (e.g., 4-month-old, 1 y 7 m cases) [H, PMID: 35170241; PMID: 37532502]. Acute seizure presentation on a chronic metabolic/developmental background.
- **Progression:** hypomagnesemia chronic/lifelong; intellectual disability generally static but permanent; epilepsy ranges controllable→intractable.

- **Course:** chronic, lifelong; no spontaneous remission of the genetic defect; seizures may be controlled with Mg^{2+} correction + antiseizure medication.
- **Critical period:** early neurodevelopment (prenatal/infancy) is the key window; early Mg^{2+} correction may limit neurological sequelae (inferred from analogous Mg^{2+}-wasting disorders, PMID 32369769).

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9. Inheritance and Population

- **Epidemiology:** very rare; ~24 pediatric cases by 2022 [H, PMID: 35170241], largest cohort ~34 [PMID: 33600043]. No reliable prevalence/incidence (ultra-rare, <1/1,000,000 order of magnitude).
- **Inheritance:** predominantly **autosomal dominant**, frequently **de novo** heterozygous variants [H, PMID: 21397062; PMID: 37532502; PMID: 35170241]; severe early-onset forms with brain malformation are **autosomal recessive** (biallelic): *"For the first time, a recessive mode of inheritance of CNNM2 mutations was observed. Importantly, patients with recessive CNNM2 mutations suffer from brain malformations and severe intellectual disability"* [H, PMID: 24699222].
- **Genotype–phenotype:** allelic-dose gradient — biallelic/recessive → severe ID + brain malformation; monoallelic *de novo* missense → moderate ID.
- **Penetrance/expressivity:** high but **variable expressivity** (severity and even presence of hypomagnesemia vary) [PMID: 33600043; PMID: 34490037].
- **Anticipation:** not applicable (not a repeat-expansion disorder).
- **Germline mosaicism / founder effects:** not established. **Consanguinity** relevant for recessive forms.
- **Carrier frequency / constraint:** no established carrier frequency; CNNM2 highly LoF-intolerant (pLI ≈ 1.0, LOEUF = 0.29), so truncating variants are very rare in the general population [C, gnomAD v2.1.1].
- **Population/sex/geography:** cases reported worldwide (Europe, China, etc.); no strong ethnic predilection or established sex bias.

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10. Diagnostics

- **Laboratory:** low **serum magnesium** (<0.7 mmol/L; LOINC 19123-9); inappropriately high **urinary Mg^{2+} / FE-Mg^{2+}** indicating renal wasting; typically normal/variable serum calcium and PTH (distinguishing from TRPM6/FHSH which causes secondary hypocalcemia, PMID 32369769).
- **Imaging:** brain MRI may show developmental/structural abnormalities in a subset [PMID: 34490037].
- **Electrophysiology:** EEG for seizure characterization; ECG may show arrhythmia/QT changes with severe hypomagnesemia.
- **Genetic testing (gold standard):** trio **whole-exome sequencing (WES)**, confirmed by Sanger [H, PMID: 37532502; PMID: 35170241]; targeted hypomagnesemia **gene panels** (CNNM2, TRPM6, FXYD2, KCNA1, EGF, SLC12A3, CLDN16/19, HNF1B/PCBD1) and single-gene CNNM2 testing are alternatives. Functional assays (Mg^{2+} flux, TRPM7 regulation, Western/immunofluorescence) support novel-variant interpretation [PMID: 40612795].
- **Differential diagnosis:** other hereditary hypomagnesemias — TRPM6 (FHSH with secondary hypocalcemia, PMID 32369769), CNNM4 (Jalili syndrome), FXYD2, EGF, KCNA1, Gitelman (SLC12A3), Claudin-16/19 (FHHNC), HNF1B/PCBD1; plus non-genetic causes (PPIs, diuretics, alcohol, GI losses, vitamin-D-deficient hypocalcemic seizures [PMID 41869998]) [H, PMID: 38871680].
- **Screening:** cascade genetic testing of relatives once a familial variant is identified; no population newborn screening.

11. Outcome / Prognosis

- **Survival/mortality:** generally not life-limiting if hypomagnesemia is controlled; risk from severe acute hypomagnesemia (seizures, arrhythmia) [H, PMID: 21397062]. Homozygous loss is embryonic-lethal in mice [PMID: 33859252].
- **Morbidity/function:** lifelong intellectual disability, epilepsy, and speech/motor impairment dominate the disability burden; chronic reliance on Mg^{2+} supplementation.
- **Complications:** intractable epilepsy, cardiac arrhythmia, GI intolerance of oral Mg^{2+}.
- **Recovery potential:** biochemical correction is achievable but **established neurodevelopmental deficits are not reversible**.

- **Prognostic factors:** genotype (allelic dose), seizure intractability, degree of developmental delay, presence of brain malformation, and timeliness of Mg^{2+} correction. Neurological outcome does not strictly correlate with serum Mg^{2+} [PMID: 34490037].

12. Treatment

No curative therapy exists; management is symptomatic/supportive.

- **Magnesium supplementation** (NCIT:C29236 Magnesium): oral Mg^{2+} salts (magnesium oxide/sulfate) first-line; renal wasting makes normalization difficult and high doses cause dose-limiting **diarrhea**. Severe/refractory cases require **IV magnesium sulfate**, and continuous **subcutaneous Mg^{2+} infusion** has normalized levels and improved symptoms in refractory renal-wasting hypomagnesemia: *"achieved near-normal serum magnesium levels with subcutaneous magnesium infusions"* [H, PMID: 36941847]. CNNM2 hypomagnesemia is typically only partially Mg^{2+}-responsive.
- **Antiseizure medication** (NCIT:C264 Anticonvulsant Agent) for epilepsy, alongside Mg^{2+} correction.
- **Potassium-sparing diuretics** (e.g., amiloride) are used in some renal Mg^{2+}-wasting disorders to reduce urinary losses (adjunct; limited CNNM2-specific evidence).
- **Supportive/rehabilitative:** developmental/educational support, speech therapy, physiotherapy, occupational therapy; cardiology follow-up where CHD/arrhythmia is present.
- **Advanced/experimental therapeutics (gene, cell, RNA, targeted):** none in clinical use; no CNNM2-specific trials. Emerging general hypomagnesemia approaches (prebiotic fibers, SGLT2 inhibitors) are **not validated for HOMGSMR1** [PMID: 38871680]. Pharmacogenomics: not applicable.

13. Prevention

- **Primary prevention:** not possible (genetic); genetic counseling for families with a known variant.

- **Secondary prevention:** early recognition—*"We suggest testing for CNNM2 mutation in patients with neurological impairment and hypomagnesemia"* [PMID: 34515155]—and early Mg^{2+} correction to limit neurological sequelae.
- **Tertiary prevention:** ongoing Mg^{2+} maintenance and seizure control to prevent complications.
- **Reproductive options:** carrier/variant testing, prenatal diagnosis, preimplantation genetic testing where a familial variant is known.
- **Public-health/immunization/environmental measures:** not applicable.

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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** CNNM2 is evolutionarily conserved [V, PMID: 24706765]. Orthologs: mouse *Cnnm2* (NCBI Taxon 10090), rat *Cnnm2*, zebrafish *cnnm2* (NCBI Taxon 7955). The CNNM family is ancient/conserved ("Ancient Conserved Domain Protein").
- **Natural disease in animals:** no well-documented spontaneous CNNM2 disease in companion animals/wildlife (no established OMIA entry); ruminant hypomagnesemia ("grass tetany") is nutritional, not CNNM2-genetic. Hypomagnesemia from other causes occurs (e.g., refeeding syndrome in a cat, PMID 35818515).
- **Comparative biology:** conserved Mg^{2+}-handling mechanism across CNNM1–4 paralogs; CNNM4 defects cause Jalili syndrome, underscoring family-wide conservation. **No zoonotic potential** (non-infectious).

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15. Model Organisms

Model	Genotype	Key phenotype	Recapitulation	Reference
Mouse (*Cnnm2*)	Homozygous KO	**Embryonic lethal**; developmental defects	Confirms essentiality	PMID: 33859252
Mouse	Systemic heterozygous KO	**Hypomagnesemia**; viable	Models dominant human disorder	PMID: 28033128
Mouse	Kidney-specific homozygous KO	Hypomagnesemia; basolateral DCT localization; reduced BP (SBP ~121.7 WT vs ~110 mmHg)	Confirms renal DCT role	PMID: 28033128
Zebrafish (*cnnm2*)	Morpholino knockdown	Disturbed brain development, increased spontaneous contractions, weak touch-evoked escape, reduced body Mg^{2+}	Models neuro + Mg^{2+} phenotype	PMID: 24699222
Zebrafish (*cnnm2*)	Orthologue mutation (schizophrenia-gene screen)	Altered forebrain development, decreased prepulse inhibition	Supports neurodevelopmental role	PMID: 30929901
HEK293 / epithelial cells	CNNM2 overexpression / mutant	WT increases $^{25}Mg^{2+}$ uptake via TRPM7; mutant does not; basolateral localization; Mg^{2+}-sensitive Na^{+} currents diminished in mutant	Defines cellular mechanism	PMID: 24699222; PMID: 21397062

Recapitulation & limitations. Mouse and zebrafish models reproduce both the renal (hypomagnesemia) and neurological (brain-development) axes, strongly validating causality. Homozygous mouse lethality precludes adult homozygous CNS modeling; morpholino knockdown carries off-target caveats; cell models capture molecular transport but not cognition/seizures. **Resources:** MGI (mouse), ZFIN (zebrafish), Cellosaurus (HEK293).

Mechanistic Model / Interpretation

The evidence converges on a single, coherent **five-layer model**:

Layer	Evidence	Conclusion
Human genetics	Heterozygous *de novo* + biallelic CNNM2 variants cause disease	CNNM2 causal; allelic-dose gradient (PMID: 21397062; PMID: 24699222; PMID: 33600043)
Population genetics	pLI ≈ 1.0, LOEUF = 0.29, mis_Z = 5.77	Haploinsufficiency; dominant LoF
Structural biology	Mg-ATP drives twisted→flat CBS switch; T568I locks it	Allosteric gating defect (PMID: 25184538)
Cell biology	Basolateral DCT Mg^{2+} handling; regulates TRPM7	Transport/sensing defect (PMID: 21397062; PMID: 24699222)
Animal models	Mouse KO (hypomagnesemic/ lethal), zebrafish (brain defects)	Recapitulate renal + neuro phenotypes (PMID: 33859252; PMID: 28033128; PMID: 24699222)

A key interpretive nuance: the neurological phenotype is **partially dissociable** from systemic hypomagnesemia. Cases with brain malformation and intractable epilepsy **without** hypomagnesemia [PMID: 34490037], plus zebrafish brain-development defects, argue for a **cell-autonomous CNS requirement** for CNNM2 in addition to the downstream effects of low serum Mg^{2+}. This explains why Mg^{2+} supplementation corrects biochemistry and aids seizure control but does not reverse cognitive deficits.

Evidence Base

PMID	Title (abbrev.)	Contribution
21397062	*CNNM2 … mutated in dominant hypomagnesemia*	Establishes causality; basolateral DCT localization
24699222	*CNNM2 mutations cause impaired brain development and seizures*	Recessive inheritance; TRPM7 mechanism; zebrafish
24706765	*Mg^{2+}-dependent ATP binding to CBS domains*	CBS domains sense Mg-ATP; disease variant abolishes efflux
25184538	*Nucleotide-triggered CBS conformational change*	Twisted→flat switch; T568I mimics nucleotide-bound state
33600043	*Phenotypic/genetic spectrum (largest cohort)*	11 novel variants; phenotype spectrum
33859252	*Cnnm2 KO mice*	Hypomagnesemia, developmental defects, lethality
28033128	*Renal CNNM2 maintains blood pressure*	Kidney-specific/heterozygous KO hypomagnesemia
35170241	*Tyr189Cys case + review*	Core triad; ~24 cases; *de novo* pathogenic missense
34490037	*Brain damage/epilepsy without hypomagnesemia*	Neuro phenotype dissociable from serum Mg^{2+}
37532502	*c.1448delT case*	*De novo* truncating pathogenic variant (ACMG)
40612795	*Novel variants p.E298del, p.P360R*	Expands mutation/functional spectrum
27899452	*PRL-1–CNNM2 structure*	Regulatory PRL-CNNM interaction raises intracellular Mg^{2+}
27856537	*Phosphocysteine in PRL-CNNM pathway*	Regulation of Mg^{2+} homeostasis
22399287	*Membrane topology/processing of CNNM2*	Structure, glycosylation, ATP-binding by CBS
38871680	*Magnesium biology (review)*	DCT Mg^{2+} physiology; CNNM2 basolateral extrusion
36941847	*Subcutaneous Mg infusion*	

PMID	Title (abbrev.)	Contribution
		Treatment for refractory renal Mg^{2+} wasting
34515155	*Diagnostic dilemma case*	Recommends CNNM2 testing in neuro + hypomagnesemia
32369769	*TRPM6 variant (FHSH)*	Differential diagnosis
30929901	*Schizophrenia-gene zebrafish atlas*	cnnm2 forebrain development role
gnomAD v2.1.1	Constraint (computational)	pLI ≈ 1.0, LOEUF = 0.29, mis_Z = 5.77 → haploinsufficiency

Limitations and Knowledge Gaps

1. **Rarity limits epidemiology.** With ~24–34 reported cases, prevalence, incidence, sex ratio, penetrance, and carrier frequency cannot be precisely estimated; phenotype frequencies remain qualitative.
2. **Genotype–phenotype map is incomplete.** The dominant-vs-recessive severity gradient is established qualitatively, but systematic variant-level functional classification is lacking for most alleles.
3. **CNS mechanism is partly inferred.** The Mg^{2+}-independent neurodevelopmental role rests on zebrafish and the hypomagnesemia-independent case; direct demonstration in human neurons is limited.
4. **Haploinsufficiency vs dominant-negative** is not fully resolved for individual variants, though gene-level constraint favors haploinsufficiency.
5. **No human treatment-outcome data.** The effect of Mg^{2+} supplementation on long-term cognitive/seizure outcomes has not been systematically studied.
6. **Some citation snippets were flagged as title-based ("mismatch")** in the knowledge state (e.g.,

P 33859252

P 34490037

)

P 27856537

and should be verified against full abstracts before public-database ingestion.

Proposed Follow-up Experiments / Actions

1. **Conditional CNS-specific Cnnm2 knockout mice** (e.g., Nestin-Cre) to dissect cell-autonomous brain requirement from systemic hypomagnesemia and test whether early Mg^{2+} correction rescues neurodevelopment.
2. **Patient iPSC-derived neurons/organoids** carrying dominant (T568I, Tyr189Cys) and recessive variants to measure Mg^{2+} handling, TRPM7 regulation, and developmental defects in a human neural context.
3. **Deep mutational scanning / functional assay** of CNNM2 variants (Mg^{2+} flux, CBS-module conformation, surface expression) to build an evidence-based ACMG variant-classification map.
4. **Natural-history registry** for HSMR to quantify onset age, seizure trajectory, cognitive outcomes, and supplementation response across genotypes.
5. **Structural studies of additional pathogenic variants** (p.P360R, p.E298del) to test whether they share the T568I "flat-lock" mechanism or act via misfolding/trafficking.
6. **Verify flagged citations** and resolve ClinVar/ACMG classifications for all reported variants before knowledge-base ingestion.

Evidence source types: human clinical (case reports/cohorts), model organism (mouse, zebrafish), in vitro (HEK293, crystallography), and computational (gnomAD constraint).
